HAZAL TAŞ

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PROFESSIONAL SUMMARY

Computer Engineer and Master's student in Data Analytics & AI with practical software engineering experience at Softtech A.Ş. Specialized in designing Generative AI pipelines, Retrieval-Augmented Generation (RAG), and agentic workflows using modern LLM architectures and Python. Strong engineering foundation in asynchronous API development, database optimization, and end-to-end machine learning lifecycle management.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python (FastAPI, Flask, Pandas, NumPy, Scikit-learn), Java (Spring Boot), SQL (MySQL, PostgreSQL), Dart (Flutter), Bash, Git
- **Generative AI & LLMs:** Multimodal AI (Vision/Embeddings), Vector Search (sqlite-vec, Pinecone, ChromaDB), RAG Pipelines, Hybrid Search, LangChain, Agentic Patterns (Tool Use, Context & State Management), Prompt Engineering
- **Backend & Systems:** RESTful APIs, Swagger/OpenAPI, Asynchronous Event Loops, Thread Pooling, CI/CD (GitHub Actions), Docker Basics, Linux
- **Core Competencies:** Data Structures & Algorithms, Object-Oriented Programming (OOP), Relational Schema Design, Fault-Tolerant Architectures

EDUCATION

Steinbeis University — Berlin, Germany

Master of Science (M.Sc.) in Business Management (Specialization: Data Analytics & AI)

July 2026 – July 2028

Çukurova University — Adana, Turkey

Bachelor of Science in Computer Engineering (100% English Instruction)

Graduated: 2024

WORK EXPERIENCE

Softtech A.Ş. — Ankara, Turkey

Software Engineering Intern | June 2024 – August 2024

- **API & Backend Development:** Engineered secure, scalable RESTful backend microservices using Java and Spring Boot to support high-throughput internal data workflows.
- **API Testing & Documentation:** Leveraged Postman and Swagger/OpenAPI to test, debug, and document REST endpoints and authentication flows, ensuring clean cross-platform integration.
- **Data & Schema Optimization:** Designed and indexed relational database schemas in MySQL, enhancing query latency and data integrity for master records.
- **Cross-Platform Integration:** Built responsive mobile client modules using Flutter, integrating them with backend APIs and structured JSON payloads.

SELECTED AI & SOFTWARE PROJECTS

AI-Powered Smart Photo Memory Assistant | Python, FastAPI, SQLite (sqlite-vec), Gemini Vision & Embeddings, React, Twilio

- Engineered a multimodal semantic retrieval system analyzing visual inputs via Gemini Vision API and vectorizing metadata using 3072-dimensional embeddings for low-latency cosine similarity search.
- Designed a hybrid search parser decomposing unstructured natural language queries into deterministic SQL filters and semantic vector searches to eliminate retrieval hallucinations.
- Built asynchronous API pipelines using FastAPI and worker thread pooling (run_in_threadpool) to eliminate event-loop blocking during concurrent high-volume media ingestion.
- Implemented production-grade fault-tolerance mechanisms, including exponential backoff and automated retry routines to handle external API rate limits (HTTP 429/503).

Automated Job Application & Pipeline Agent | Python, Gemini API, GitHub Actions, Google Sheets API, Telegram Bot

- Developed an autonomous end-to-end job tracking and parsing agent scraping real-time postings across European tech job boards twice daily.
- Integrated Gemini LLM workflows to dynamically parse job descriptions, generate tailored cover letters, and produce ATS-optimized CV drafts based on role criteria.
- Orchestrated scheduled automated pipelines via GitHub Actions, persisting structured pipeline states and application metrics directly into live spreadsheets and Telegram alerts.

AI Tandem – Conversational Language Partner | Python, LLM APIs, Prompt Engineering, Speech/NLP Pipelines

- Created an interactive conversational AI application designed for real-time language practice, adaptive dialogue generation, and context-aware feedback.
- Implemented stateful context management and prompt orchestration to dynamically assess user input, correct grammatical nuances, and adjust conversational complexity.